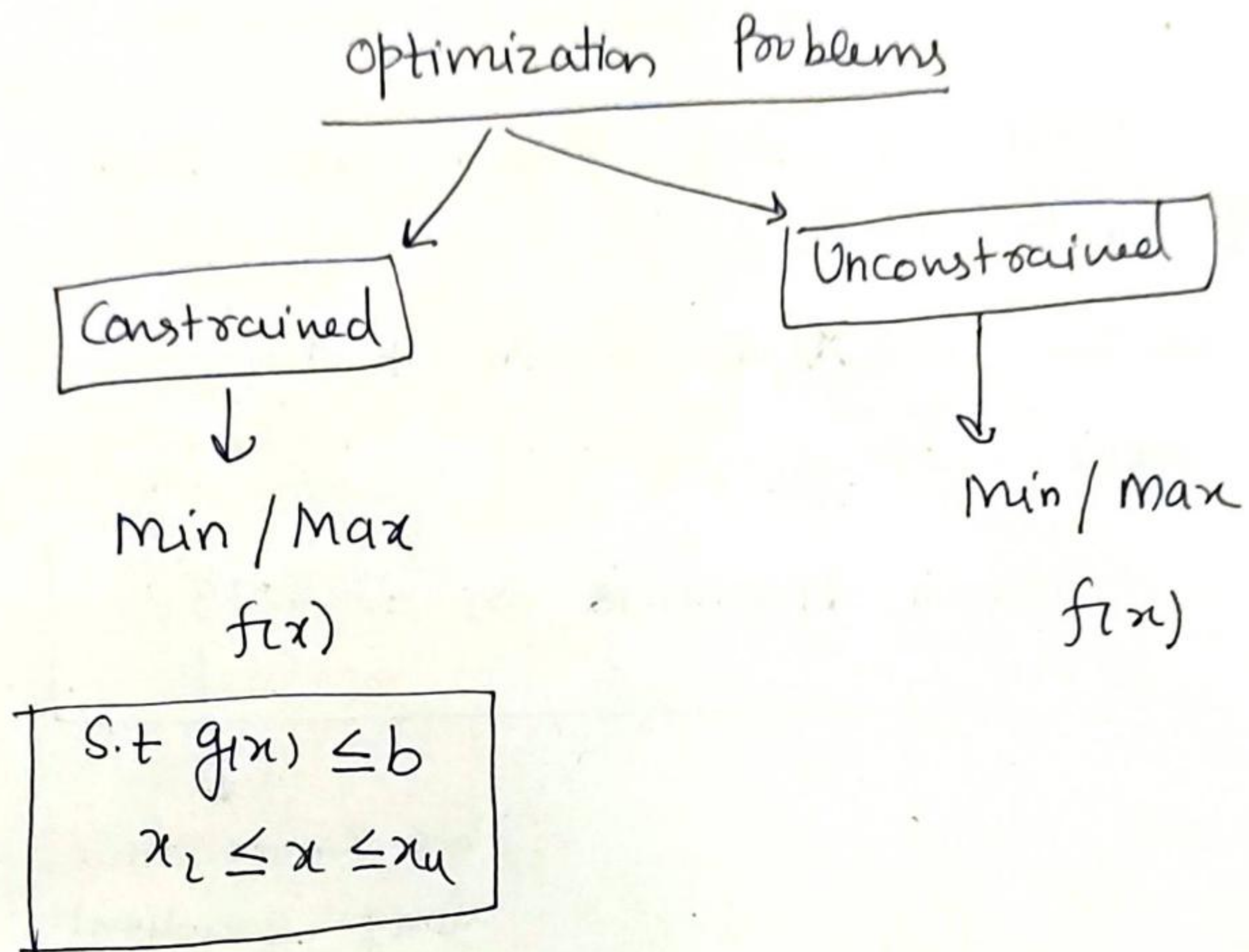


①

Steepest Descent Method (Cauchy) Method
 (for unconstrained optimization problems)
 Gradient Based Technique



Unconstrained Minimization Methods

Direct Search Methods (Zeroth order method) (No derivative)	Descent Methods (Gradient method) ↗
Random search method	Steepest Descent (Cauchy) Method
Univariate method	Fletcher - Reeves method
Pattern search method	Newton method
Powell's method	Quasi-Newton method
Hooke - Jeeves method	Davidson - Fletcher - Powell method
Simplex method	...

General strategy for Gradient methods:-

Minimize $f(x)$

- Step 1:- Calculate the search direction S_i .
- Step 2:- Select a step length in that direction to reduce $f(x)$

$$X_{i+1} = X_i + \lambda S_i$$

Steepest Descent:-

Search Direction is $S_i = -\nabla f_i$

We know that, if we move along gradient direction from any point to n -dimensional space, the function value increases at the fastest rate.

Gradient Descent (Cauchy) Method:-

- It uses the negative of the gradient vector as the direction for minimization. $S_i = -\nabla f$
- Start from the initial point X_1 and iteratively move along the steepest descent directions until the optimum point is reached.

Algorithm Steepest Descent:-

Minimize $f(x)$

Max $\rightarrow$ Min
 $f \rightarrow -f$

Minimize $f(x)$

Start the arbitrary x_1 , Set Iteration number $i=1$.

Step 1:- Find the search direction s_i as

$$s_i = -\nabla f_i = -\nabla f(x_i)$$

Step 2:- Determine the optimal step length d_i in the direction of s_i and set

$$x_{i+1} = x_i + d_i s_i$$

Step 3:- Test the optimality for the new point x_{i+1} .

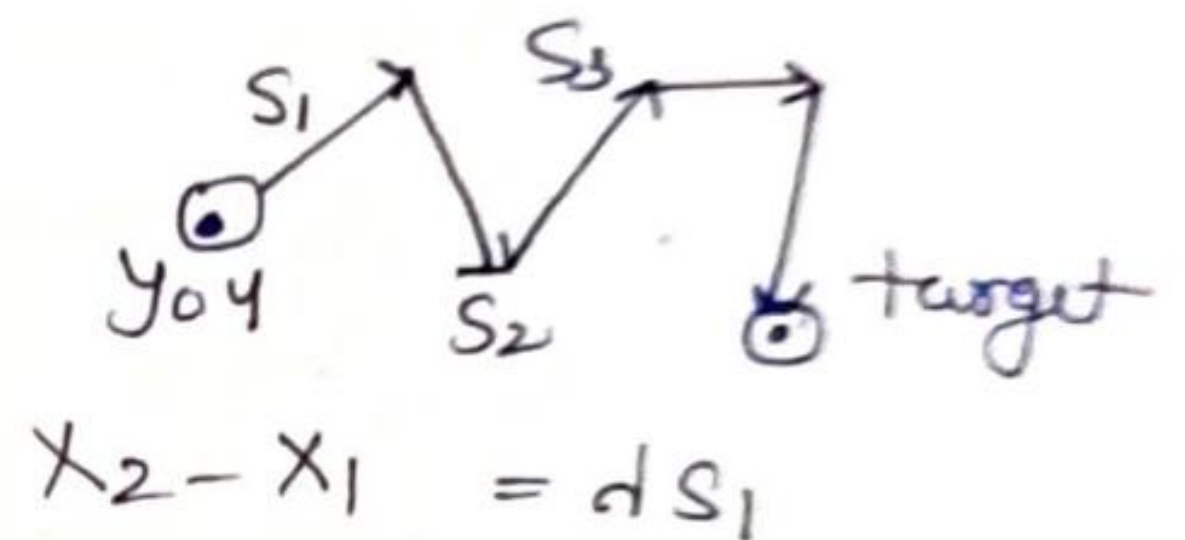
How to pick d ?

Analytically

Numerically

Bisection
Newton
Secant

Task of d is to minimize a function in search direction s .



Numerical method:-

Using searching technique to
Fixed ($d=1$) or Variable ($d=1, 2, \frac{1}{2}, \text{etc}$)

options for optimizing d:-

- 1) Use interpolation such as quadratic, cubic
- 2) Region elimination (Golden Search)
- 3) Newton, Secant, Quasi-Newton
- 4) Random
- 5) Analytical optimization

(1), (3) and (5) are preferred. However it may be desirable to exactly optimize d (better to generate new search directions).

Analytic method:-

How does one minimize a function in a search direction using an analytic method?
It means S is fixed and you want to pick d , the step length to minimize $f(x)$.
Note $\Delta x_i = d S_i$.

$$\begin{aligned} f(x_{i+1}) &= f(x_i + d S_i) \\ &\approx f(x_i) + \nabla^T f(x_i) (\Delta x_i) + \frac{1}{2} (\Delta x_i)^T H(x_i) (\Delta x_i) \end{aligned}$$

Thus,

$$\begin{aligned} \frac{df(x_i + d S_i)}{dd} &= 0 \\ \Rightarrow \nabla f_i^T S_i + d (S_i)^T H_i S_i &= 0 \\ \Rightarrow d &= - \frac{\nabla f_i^T S_i}{S_i^T H_i S_i} = \frac{S_i^T S_i}{S_i^T H_i S_i} \end{aligned}$$

This yields a minimum of the approximating function

of Steepest of Descent method to minimize the function $f(x)$

(3 steps rules to solve the problems)

Steps for Numerical example:-

Step 1:- Calculate S_i at X_i by $S_i = -\nabla f_i$ (search direction)

Step 2:- Calculate d_i by using $d_i = \frac{S_i^T S_i}{S_i^T H_i S_i}$ (step length) and the new point

$$X_{i+1} = X_i + d_i S_i$$

Step 3:- check the optimum of X_{i+1} by $\nabla f(X_{i+1}) \cong 0$.



If met stop, otherwise implement step 1 again for this new point X_{i+1}

Sol:-

Minimize

$$f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$$

starting from the point $X_1 = (0, 0)$

Sol:-

The gradient of f is

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

$$\nabla f_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Hessian matrix is

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$
$$= \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

Iteration 1:-

At $X_1 = (0, 0)$

Step 1:-

Find S_1 at X_1

$$S_1 = -\nabla f(X_1) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Step 2:-

Compute d_1 at X_1

$$d_1 = \frac{S_1^T S_1}{S_1^T H S_1} = \frac{2}{2} = 1$$

$$= \frac{[-1, 1] \begin{bmatrix} -1 \\ 1 \end{bmatrix}}{[-1, 1] \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}} = \frac{-2}{[-1, 1] \begin{bmatrix} -2 \\ 0 \end{bmatrix}}$$

$$= \frac{-2}{+2} = 1$$

Hence the new point is

$$X_2 = X_1 + d_1 S_1$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Step 3:-

check the optimum

$$\nabla f(X_2) = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So X_2 is not optimum. So move next iteration.

Iteration 4:- At x_4 :-

Step 1:- $S_4 = -\nabla f(x_4) = \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix}$

Step 2:- $d_4 = \frac{S_4^T S_4}{S_4^T H S_4} = \frac{1}{5}$

Hence, the new point is

$$\begin{aligned} x_5 &= x_4 + d_4 S_4 \\ &= \begin{bmatrix} -1.0 \\ 1.4 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} -0.96 \\ 1.44 \end{bmatrix} \end{aligned}$$

Step 3:- Since $\nabla f(x_5) = \begin{bmatrix} 0.04 \\ -0.04 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

So, x_5 is optimum.

This process has to be continued until the optimum point

$$x = \begin{bmatrix} -1.0 \\ 1.5 \end{bmatrix} \text{ is found.}$$

iteration 5 and it 6

$$\boxed{d = 0}$$

Iteration 2:- $Ax_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Step 1:- $S_2 = -\nabla f(x_2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Step 2:- $d_2 = \frac{S_2^T S_2}{S_2^T H S_2} = \frac{2}{10} = \frac{1}{5}$

Hence, the new point is

$$\begin{aligned} x_3 &= x_2 + d_2 S_2 \\ &= \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} \end{aligned}$$

Step 3:-

Since $\nabla f(x_3) = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

So, x_2 is not optimum. So move next iteration.

Iteration 3:-

Ax_3

Step 1:- $S_3 = -\nabla f(x_3) = \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$

Step 2:- $d_3 = \frac{S_3^T S_3}{S_3^T H S_3} = 1$

Hence, the new point is

$$\begin{aligned} x_4 &= x_3 + d_3 S_3 \\ &= \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} + 1 \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} -1.0 \\ 1.4 \end{bmatrix} \end{aligned}$$

Step 3:-

Since $\nabla f(x_4) = \begin{bmatrix} -0.2 \\ -0.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

So, x_4 is not optimum. So we move next iteration.

Min $f(x_1, x_2) = 4x_1^2 - 4x_1x_2 + 2x_2^2$
 Starting from the point $x_1 = (2, 3)$. (2nd iteration)

Sol:-

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 8x_1 - 4x_2 \\ -4x_1 + 4x_2 \end{bmatrix}$$

$$\nabla f_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 16 - 12 \\ -8 + 12 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$S_1 = -\nabla f_1 = \begin{bmatrix} -4 \\ -4 \end{bmatrix}$$

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix}$$

$$d_1 = \frac{S_1^T S_1}{S_1^T H S_1}$$

$$d_1 = \frac{\begin{bmatrix} -4 & -4 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \end{bmatrix}}{\begin{bmatrix} -4 & -4 \end{bmatrix} \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} -4 \\ -4 \end{bmatrix}}$$

$$d_1 = \frac{32}{\begin{bmatrix} -4 & -4 \end{bmatrix} \begin{bmatrix} -32 + 16 \\ 16 - 16 \end{bmatrix}}$$

$$d_1 = \frac{32}{\begin{bmatrix} -4 & -4 \end{bmatrix} \begin{bmatrix} -16 \\ 0 \end{bmatrix}}$$

$$d_1 = \frac{32}{64} = \frac{1}{2}$$

$$\begin{aligned}
 x_2 &= x_1 + d_1 s_1 \\
 &= \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} -4 \\ -4 \end{bmatrix} \\
 &= \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ -2 \end{bmatrix} \\
 x_2 &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}
 \end{aligned}$$

$$\nabla f_2 = \begin{bmatrix} 8x_0 - 4x_1 \\ -4x_0 + 4x_1 \end{bmatrix}$$

$$\nabla f_2 = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$$

$$s_2 = -\nabla f_2 = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

$$\begin{aligned}
 d_2 &= \frac{s_2^T s_2}{s_2^T H_2 s_2} \\
 &= \frac{\begin{bmatrix} 4, -4 \end{bmatrix} \begin{bmatrix} 4 \\ -4 \end{bmatrix}}{\begin{bmatrix} 4, -4 \end{bmatrix} \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -4 \end{bmatrix}}
 \end{aligned}$$

$$= \frac{32}{\begin{bmatrix} 4, -4 \end{bmatrix} \begin{bmatrix} 48 \\ -32 \end{bmatrix}} = \frac{32}{48 \times 4 + 4 \times 32} = \frac{1}{10} = 0.1$$

$$\begin{aligned}
 x_3 &= x_2 + d_2 s_2 \\
 &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0.1 \begin{bmatrix} 4 \\ -4 \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}
 \end{aligned}$$